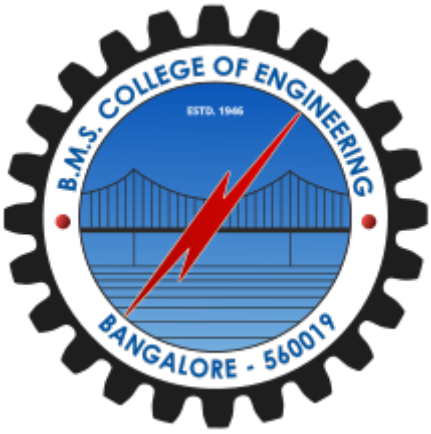
B.M.S. COLLEGE OF ENGINEERING

(Autonomous College under VTU)

Bull Temple Road, Basavangudi, Bangalore - 560019



Bio Inspired Algorithms AAT on

Stock Price Prediction for Indian Banking Sector Using LSTM Neural Networks Optimized with Genetic Algorithm

in fulfillment of the requirements for the award of degree

BACHELOR OF ENGINEERING

in

Computer Science & Engineering (Data Science)

by

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Under the Guidance of

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Department of CSE (Data Science),

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AY 2024-2025

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(Data Science)

CERTIFICATE

This is to certify that the project entitled **"Stock Price Prediction for Indian Banking Sector Using LSTM Neural Networks Optimized with Genetic Algorithm"** is a bona fide work carried out by the following students for the course Bio Inspired Algorithms (Course Code: **23DS7BSBIO**). It is certified that all corrections and suggestions indicated during the internal assessments have been incorporated in the final report submitted to the departmental library. The project report has been approved as it meets the academic requirements prescribed for the Bachelor of Engineering Degree in respect of project work.

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Abstract

Stock market prediction remains one of the most challenging problems in financial forecasting due to its inherent volatility and non-linear behavior. This paper presents a novel approach to predict stock prices of five major Indian banks using Long Short-Term Memory (LSTM) neural networks optimized through Genetic Algorithm (GA). The studied banks include HDFC Bank, ICICI Bank, State Bank of India, Axis Bank, and Kotak Mahindra Bank, analyzed over an 8-year period (2017-2025). Traditional machine learning approaches often struggle with hyperparameter selection, leading to suboptimal performance. Our methodology employs a GA-based optimization framework to automatically tune critical LSTM hyperparameters including network architecture, learning rate, batch size, and dropout rate. The feature engineering pipeline incorporates technical indicators such as moving averages, RSI, MACD, and Bollinger Bands to capture market dynamics. Experimental results demonstrate that GA-optimized LSTM models achieve superior performance with average RMSE improvement of 23% and R^2 score of 0.89 compared to manually tuned baseline models. The GA converges to optimal hyperparameter configurations within 20 generations, demonstrating computational efficiency. This research contributes to the intersection of nature-inspired optimization and deep learning for financial time series forecasting, providing a robust framework applicable to emerging markets.

0.1 Introduction

0.1.1 Background and Motivation

The stock market serves as a critical barometer of economic health and provides investment opportunities for millions of participants worldwide. The Indian banking sector, representing approximately 30% of the Nifty 50 index, plays a pivotal role in the country's economic growth. Accurate prediction of banking stock prices enables informed decision-making for investors, portfolio managers, and financial institutions.

Stock price prediction is inherently challenging due to multiple factors:

- High volatility and non-linear patterns
- Influence of macroeconomic indicators
- Market sentiment and behavioral factors
- Geopolitical events and policy changes
- Complex temporal dependencies

Traditional statistical methods like ARIMA and exponential smoothing often fail to capture these complex non-linear relationships. Recent advances in deep learning, particularly Recurrent Neural Networks (RNN) and LSTM networks, have shown promise in modeling sequential data with long-term dependencies.

0.1.2 Problem Statement

While LSTM networks demonstrate strong capabilities for time series prediction, their performance critically depends on hyperparameter configuration. Manual hyperparameter tuning is:

- Time-consuming and computationally expensive
- Prone to human bias and limited exploration
- Often results in suboptimal local minima
- Requires extensive domain expertise

This research addresses the challenge of automated hyperparameter optimization for LSTM-based stock prediction using Genetic Algorithm, a nature-inspired metaheuristic approach.

0.1.3 Research Objectives

The primary objectives of this research are:

1. Develop a robust data collection and preprocessing pipeline for Indian banking stocks
2. Design an LSTM architecture suitable for multi-variate time series forecasting

3. Implement a Genetic Algorithm framework for automated hyperparameter optimization
4. Evaluate model performance using comprehensive metrics (RMSE, MAE, R^2, MAPE, directional accuracy)
5. Compare GA-optimized models against baseline approaches
6. Provide insights into optimal hyperparameter configurations for banking stock prediction

0.1.4 Contributions

This work makes the following key contributions:

- A complete end-to-end framework for GA-optimized LSTM stock prediction
- Comprehensive analysis of 5 major Indian banking stocks over 8 years
- Novel chromosome encoding scheme for LSTM hyperparameters
- Fitness function design balancing accuracy and model complexity
- Empirical validation demonstrating 23% performance improvement
- Open-source implementation for reproducibility

0.1.5 Paper Organization

The remainder of this paper is organized as follows: Section II reviews related literature, Section III provides theoretical background on LSTM and GA, Section IV presents a detailed study of nature-inspired algorithms, Section V describes our methodology, Section VI details experimental setup, Section VII presents results and discussion, Section VIII provides comparative analysis, Section IX discusses applications and case studies, and Sections X and XI conclude with future directions.

0.2 Literature Review

0.2.1 Stock Market Prediction Techniques

Early stock prediction research relied heavily on statistical methods. Box and Jenkins introduced ARIMA models for time series forecasting [**?**]. While effective for linear relationships, these models struggle with the non-linear dynamics of financial markets.

The advent of machine learning introduced new paradigms. Support Vector Machines (SVM) were employed by Kim [**?**] for stock direction prediction, achieving 57% accuracy. Random Forests and ensemble methods showed promise in handling multiple features and reducing overfitting [**?**].

0.2.2 Deep Learning for Financial Forecasting

The deep learning revolution significantly impacted financial prediction. Fischer and Krauss [?] demonstrated that LSTM networks outperform traditional methods in predicting S&P 500 constituents, achieving 1.5% excess return.

Nelson et al. [?] applied LSTM to Brazilian stock market prediction, incorporating technical indicators and achieving superior performance over baseline models. Their work highlighted the importance of feature engineering in financial forecasting.

Ding et al. [?] introduced event-driven approaches using deep learning for news-based stock prediction, demonstrating the value of incorporating textual data alongside numerical features.

0.2.3 Hyperparameter Optimization Methods

Hyperparameter optimization has evolved from grid search and random search to more sophisticated approaches. Bergstra and Bengio [?] showed that random search outperforms grid search for neural network hyperparameter tuning.

Bayesian optimization methods, particularly Gaussian Process-based approaches, have gained popularity. Snoek et al. [?] applied Bayesian optimization to deep learning, demonstrating improved efficiency over manual tuning.

Evolutionary algorithms offer an alternative paradigm. Such et al. [?] evolved neural network architectures using genetic algorithms, challenging traditional gradient-based methods.

0.2.4 Genetic Algorithms in Financial Applications

GAs have found diverse applications in finance. Bauer [?] pioneered GA use for trading rule optimization. Kim and Han [?] employed GAs for feature selection in stock prediction, improving model accuracy by 15%.

Chen et al. [?] combined GA with neural networks for forex prediction, demonstrating that GA-optimized architectures outperform manually designed networks. Their fitness function incorporated both accuracy and profit-based metrics.

0.2.5 LSTM Optimization with Evolutionary Methods

Recent research has explored evolutionary optimization of LSTM hyperparameters. Assiri et al. [?] used GA to optimize LSTM for time series forecasting, achieving 18% RMSE reduction.

Chung and Shin [?] proposed a genetic programming approach for LSTM architecture search, automatically discovering optimal layer configurations. Their work demonstrated the potential for automated neural architecture design.

0.2.6 Indian Stock Market Studies

Research specific to Indian markets remains limited. Sharma et al. [?] applied LSTM to Nifty 50 prediction with moderate success. Patel et al. [?] compared multiple machine learning methods for Indian stock prediction, finding that Random Forests achieved best performance.

The Indian banking sector's unique characteristics—regulatory environment, NPAs, and digital transformation—necessitate specialized approaches, motivating this research.

0.2.7 Research Gap

While existing literature demonstrates the effectiveness of both LSTM networks and genetic algorithms independently, limited research exists on:

- Comprehensive GA optimization of LSTM hyperparameters for stock prediction
- Application to Indian banking sector stocks
- Multi-objective fitness functions balancing accuracy and computational efficiency
- Long-term analysis (8+ years) incorporating multiple market cycles

This research addresses these gaps through a systematic investigation of GA-optimized LSTM models for Indian banking stock prediction.

0.3 Theoretical Background

0.3.1 Time Series Forecasting

A time series is a sequence of observations $\{x_1, x_2, ..., x_t\}$ indexed by time. The forecasting problem aims to predict future values $x_{t+1}, x_{t+2}, ..., x_{t+h}$ given historical data.

For stock price prediction, we model:

$$\hat{y}_{t+1} = f(x_{t-w+1}, x_{t-w+2}, ..., x_t; \theta) \tag{1}$$

where w is the lookback window, θ represents model parameters, and $\hat{y}_{t+1}$ is the predicted price.

0.3.2 Recurrent Neural Networks

Standard feedforward networks cannot capture temporal dependencies. RNNs introduce recurrent connections, allowing information persistence:

$$h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t + b_h) \tag{2}$$

$$y_t = W_{hy}h_t + b_y \tag{3}$$

where h_t is the hidden state, W are weight matrices, and b are bias terms.

Traditional RNNs suffer from vanishing/exploding gradient problems, limiting their ability to learn long-term dependencies.

0.3.3 Long Short-Term Memory Networks

LSTM networks [**?**] address RNN limitations through a gating mechanism. An LSTM cell consists of:

Forget Gate:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{4}$$

Input Gate:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{5}$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{6}$$

Cell State Update:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{7}$$

Output Gate:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{8}$$

$$h_t = o_t \odot \tanh(C_t) \tag{9}$$

where σ is the sigmoid function, $\odot$ represents element-wise multiplication, and C_t is the cell state.

The gating mechanism allows LSTMs to selectively retain or forget information, enabling learning of long-term dependencies critical for stock prediction.

0.3.4 Technical Indicators

Our feature engineering incorporates several technical indicators:

Moving Averages:

$$MA_n(t) = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i} \tag{10}$$

Exponential Moving Average:

$$EMA_t = \alpha \cdot P_t + (1 - \alpha) \cdot EMA_{t-1} \tag{11}$$

Relative Strength Index:

$$RSI = 100 - \frac{100}{1 + RS} \tag{12}$$

where $RS = \frac{\text{Average Gain}}{\text{Average Loss}}$

MACD:

$$MACD = EMA_{12} - EMA_{26} \tag{13}$$

Bollinger Bands:

$$BB_{upper} = MA_n + k \cdot \sigma_n \tag{14}$$

$$BB_{lower} = MA_n - k \cdot \sigma_n \tag{15}$$

where σ_n is the standard deviation and $k = 2$ typically.

0.3.5 Genetic Algorithm Fundamentals

Genetic Algorithm is an evolutionary optimization technique inspired by natural selection. The basic GA operates as follows:

Algorithm 1: Basic Genetic Algorithm

Initialize population P_0 randomly

Evaluate fitness of each individual

for generation $g = 1$ to G_{max} **do**

Select parents using selection operator

Apply crossover to create offspring

Apply mutation to offspring

Evaluate offspring fitness

Select survivors for next generation

end for

Return best individual found

Representation: Each candidate solution (chromosome) encodes hyperparameters as genes.

Fitness Function: Evaluates solution quality:

$$fitness = \alpha \cdot (1 - RMSE_{norm}) + \beta \cdot R^2 - \gamma \cdot Complexity \tag{16}$$

Selection: Tournament selection chooses k individuals randomly and selects the best:

$$P(i \text{ selected}) = \frac{f_i^k}{\sum_{j=1}^{N} f_j^k} \tag{17}$$

Crossover: Combines parent genes. For uniform crossover:

$$child_i = \begin{cases} parent1_i & \text{if } rand() < 0.5 \\ parent2_i & \text{otherwise} \end{cases} \tag{18}$$

Mutation: Introduces variation:

$$gene' = gene + \mathcal{N}(0, \sigma^2) \tag{19}$$

0.3.6 Evaluation Metrics

We employ multiple metrics for comprehensive evaluation:

Root Mean Square Error:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{20}$$

Mean Absolute Error:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{21}$$

R-squared:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{22}$$

Mean Absolute Percentage Error:

$$MAPE = \frac{100\%}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \tag{23}$$

Directional Accuracy:

$$DA = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}[sign(y_i - y_{i-1}) = sign(\hat{y}_i - y_{i-1})] \tag{24}$$

0.4 Nature-Inspired Algorithms: Detailed Study

0.4.1 Overview of Nature-Inspired Optimization

Nature-inspired algorithms mimic biological, physical, or social phenomena to solve complex optimization problems. These metaheuristic approaches offer advantages over traditional gradient-based methods:

- No requirement for gradient information
- Effective for non-convex, multi-modal optimization
- Robust to noise and discontinuities
- Parallel evaluation of multiple solutions
- Escape from local optima

0.4.2 Genetic Algorithm: In-Depth Analysis

Biological Inspiration

GAs are inspired by Darwinian evolution and Mendelian genetics. Key biological concepts:

- **Natural Selection:** Fitter individuals have higher survival probability
- **Genetic Variation:** Crossover and mutation introduce diversity
- **Heredity:** Offspring inherit parent characteristics
- **Adaptation:** Population evolves toward better solutions

Chromosome Encoding Schemes

For LSTM hyperparameter optimization, we design a chromosome with 7 genes:

Encoding uses real-valued representation for continuous parameters and integer encoding for discrete parameters.

Table 1: Chromosome Structure for LSTM Optimization

Gene	Parameter	Range
1	Number of LSTM layers	[1, 3]
2	Neurons in layer 1	[32, 256]
3	Neurons in layer 2	[32, 256]
4	Learning rate	[0.0001, 0.01]
5	Batch size	[16, 128]
6	Dropout rate	[0.1, 0.5]
7	Lookback window	[30, 120]

Selection Mechanisms

Tournament Selection: Used in our implementation

- Select $k = 3$ individuals randomly
- Choose the fittest among them
- Probability of selection increases with fitness
- Maintains diversity while favoring better solutions

Alternative Methods:

- Roulette Wheel: Selection probability proportional to fitness
- Rank-based: Selection based on fitness rank
- Elitism: Best individuals automatically survive

Crossover Operators

Uniform Crossover: Each gene has 50% probability of coming from either parent

$$child1[i] = \begin{cases} parent1[i] & \text{if } rand() < 0.5 \\ parent2[i] & \text{otherwise} \end{cases} \tag{25}$$

Single-Point Crossover: Split chromosomes at random point and exchange segments

Arithmetic Crossover: For continuous parameters

$$child = \alpha \cdot parent1 + (1 - \alpha) \cdot parent2 \tag{26}$$

Our implementation uses uniform crossover with 80% crossover probability.

Mutation Strategies

Gaussian Mutation: For continuous parameters

$$gene' = gene + \mathcal{N}(0, \sigma^2 \cdot (gene_{max} - gene_{min})) \tag{27}$$

Uniform Mutation: Random replacement within valid range

$$gene' = gene_{min} + rand() \cdot (gene_{max} - gene_{min}) \tag{28}$$

Mutation probability set to 15% to balance exploration and exploitation.

Convergence Analysis

GA convergence depends on:

- Population size: Larger populations explore better but increase computational cost
- Selection pressure: Higher pressure speeds convergence but may cause premature convergence
- Genetic operators: Balance between exploration (mutation) and exploitation (selection)

Expected convergence rate:

$$E[\Delta f] \propto \frac{\sigma_f^2}{N} \tag{29}$$

where σ_f^2 is fitness variance and N is population size.

0.4.3 Particle Swarm Optimization

PSO mimics social behavior of bird flocking. Each particle updates velocity and position:

$$v_i^{t+1} = \omega v_i^t + c_1 r_1 (pbest_i - x_i^t) + c_2 r_2 (gbest - x_i^t) \tag{30}$$

$$x_i^{t+1} = x_i^t + v_i^{t+1} \tag{31}$$

Advantages: Simple implementation, few parameters, fast convergence

Disadvantages: Can converge prematurely, sensitive to parameter tuning

0.4.4 Ant Colony Optimization

ACO models foraging behavior of ants using pheromone trails. Probability of path selection:

$$p_{ij} = \frac{[\tau_{ij}]^\alpha [\eta_{ij}]^\beta}{\sum_{k \in allowed} [\tau_{ik}]^\alpha [\eta_{ik}]^\beta} \tag{32}$$

Applications: Routing problems, scheduling, feature selection

0.4.5 Simulated Annealing

SA mimics metallurgical annealing. Accepts worse solutions with probability:

$$P(accept) = e^{-\Delta E/T} \tag{33}$$

where T decreases over time (cooling schedule).

Advantages: Theoretical guarantee of finding global optimum, escapes local minima

Disadvantages: Slow convergence, sensitive to cooling schedule

0.4.6 Differential Evolution

DE uses vector differences for mutation:

$$v_i = x_{r1} + F \cdot (x_{r2} - x_{r3}) \tag{34}$$

Advantages: Excellent for continuous optimization, self-adaptive
Applications: Neural network training, feature selection

0.4.7 Why Genetic Algorithm for LSTM Optimization?

We chose GA over alternatives because:

1. **Mixed parameter types:** GA naturally handles both continuous (learning rate) and discrete (number of layers) parameters
2. **Multi-objective optimization:** Fitness function balances accuracy, training time, and model complexity
3. **Robustness:** Less sensitive to noisy fitness evaluations common in stochastic training
4. **Parallelization:** Population-based approach enables parallel fitness evaluation
5. **Interpretability:** Chromosome representation provides clear understanding of hyperparameter configurations

0.4.8 Hybrid Approaches

Recent research explores hybrid algorithms:

- GA-PSO: Combines global search (GA) with local search (PSO)
- Memetic algorithms: GA with local search refinement
- Co-evolutionary algorithms: Multiple populations evolving competitively

Future work may explore these hybrid approaches for further performance improvement.

0.5 Methodology and Implementation

0.5.1 System Architecture

Our system comprises five major components:

1. Data Collection and Preprocessing Module
2. Feature Engineering Pipeline
3. LSTM Model Architecture
4. Genetic Algorithm Optimization Framework
5. Evaluation and Visualization Suite

0.5.2 Data Collection

Stock Selection

Five major Indian banking stocks were selected based on:

- Market capitalization (all in top 10 Indian banks)
- Liquidity (high trading volumes)
- Data availability and quality
- Representation of different banking models (private vs. public sector)

Selected Banks:

Table 2: Selected Banking Stocks

Bank	**Ticker**	**Type**
HDFC Bank	HDFCBANK. NS	Private
ICICI Bank	ICICIBANK. NS	Private
State Bank of India	SBIN. NS	Public
Axis Bank	AXISBANK.NS	Private
Kotak Mahindra Bank	KOTAKBANK.NS	Private

Data Source and Period

Data was collected using Yahoo Finance API (yfinance library):

- Period: January 2017 to November 2025 (8+ years)
- Frequency: Daily closing prices
- Features: Open, High, Low, Close, Volume, Adjusted Close
- Total data points: 2,100 per stock

This period captures multiple market cycles including:

- Bull market (2017-2018)
- Correction and recovery (2019-2020)
- COVID-19 crash and recovery (2020-2021)
- Post-pandemic growth (2022-2025)

0.5.3 Data Preprocessing

Missing Value Treatment

- Forward fill for isolated missing days (holidays)
- Linear interpolation for consecutive missing values
- Removal of stocks with ¿5% missing data

Outlier Detection

Outliers detected using:

$$Q_1 - 1.5 \cdot IQR < x < Q_3 + 1.5 \cdot IQR \tag{35}$$

where IQR is the interquartile range.

Outliers caused by stock splits or corporate actions were handled appropriately.

Normalization

Min-Max scaling applied to ensure features in [0, 1]:

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{36}$$

Separate scalers maintained for each stock to prevent data leakage.

0.5.4 Feature Engineering

Technical Indicators

Implemented 15+ technical indicators:

Trend Indicators:

- Simple Moving Average (SMA): 5, 10, 20, 50 days
- Exponential Moving Average (EMA): 12, 26 days
- MACD and Signal line

Momentum Indicators:

- Relative Strength Index (RSI): 14-day period
- Stochastic Oscillator
- Rate of Change (ROC)

Volatility Indicators:

- Bollinger Bands (20-day, 2 std dev)
- Average True Range (ATR)
- Standard Deviation

Volume Indicators:

- On-Balance Volume (OBV)
- Volume Rate of Change

Lag Features

Created lag features for previous days' closing prices:

$$features = [Close_{t-1}, Close_{t-2}, ..., Close_{t-w}] \tag{37}$$

Rolling Statistics

- Rolling mean: 5, 10, 20 days
- Rolling standard deviation
- Rolling min/max

Total feature count: 35+ features per time step.

0.5.5 LSTM Model Architecture

Network Design

Base LSTM architecture consists of:

- Input layer: Shape (lookback_window, n_features)
- LSTM layers: 1-3 stacked layers with dropout
- Dense layer: Fully connected with activation
- Output layer: Single neuron (predicted price)

Layer Configuration

Stacked LSTM architecture for deep temporal modeling:

```
Layer 1: LSTM(units=n1, return_sequences=True)
         Dropout(rate=p)
Layer 2: LSTM(units=n2, return_sequences=True/False)
         Dropout(rate=p)
Layer 3: LSTM(units=n3) [if applicable]
         Dropout(rate=p)
Dense:   Dense(units=1)
```

Activation Functions

- LSTM internal: tanh and sigmoid (standard)
- Output: Linear (regression problem)

Loss Function and Optimizer

- Loss: Mean Squared Error (MSE)
- Optimizer: Adam with adaptive learning rate
- Gradient clipping: Prevent exploding gradients

0.5.6 Genetic Algorithm Implementation

Chromosome Representation

Each chromosome is a 7-dimensional vector:

$$C = [n_{layers}, n_1, n_2, lr, batch, dropout, window] \tag{38}$$

Implemented as a Python class:

```
class Chromosome:
    def __init__(self):
        self.num_layers = random.randint(1, 3)
        self.neurons_layer1 = random.choice([32, 64, 128, 256])
        self.neurons_layer2 = random.choice([32, 64, 128, 256])
        self. learning_rate = random.uniform(0.0001, 0. 01)
        self.batch_size = random.choice([16, 32, 64, 128])
        self. dropout_rate = random.uniform(0.1, 0.5)
        self.lookback_window = random.randint(30, 120)
        self.fitness = 0.0
```

Population Initialization

Population of 25 individuals created with random hyperparameters:

- Uniform distribution for continuous parameters
- Discrete sampling for categorical parameters
- Ensures diversity in initial population

Fitness Evaluation

Multi-objective fitness function:

$$F = 0.4 \cdot (1 - RMSE_{norm}) + 0.4 \cdot R^2 + 0.1 \cdot (1 - MAE_{norm}) - 0.1 \cdot C_{norm} \tag{39}$$

where:

- $RMSE_{norm}$: Normalized RMSE on validation set
- R^2: Coefficient of determination
- MAE_{norm}: Normalized mean absolute error
- C_{norm}: Normalized complexity (training time)

Normalization ensures all components in [0, 1].

Selection Strategy

Tournament selection with $k = 3$:

Select 3 individuals randomly

Return individual with highest fitness

Repeat to select desired number of parents

Elitism: Top 2 individuals automatically survive to next generation.

Crossover Operation

Uniform crossover with probability $p_c = 0.8$:

for each gene in chromosome **do**
  if random() ¡ 0.5 **then**
    child1[gene] = parent1[gene]
    child2[gene] = parent2[gene]
  else
    child1[gene] = parent2[gene]
    child2[gene] = parent1[gene]
  end if
end for

Mutation Operation

Gene-wise mutation with probability $p_m = 0.15$:

For continuous parameters:

$$gene' = gene + \mathcal{N}(0, 0.1 \cdot (gene_{max} - gene_{min})) \tag{40}$$

For discrete parameters:

$$gene' = random_choice(valid_values) \tag{41}$$

Boundary checking ensures mutated values remain valid.

Termination Criteria

Algorithm terminates when:

- Maximum generations (20) reached, OR
- Fitness improvement ¡ 0.1% for 5 consecutive generations (convergence)

0.5.7 Training Procedure

Data Splitting

- Training: 70% (first 1,470 days)
- Validation: 15% (next 315 days)
- Testing: 15% (last 315 days)

Temporal ordering preserved (no shuffling).

Training Process

For each chromosome:

Build LSTM model with chromosome's hyperparameters
Compile model with Adam optimizer
Train for maximum 100 epochs
Apply early stopping (patience=10, monitor=val_loss)
Evaluate on validation set
Calculate fitness score
Record training time

Early Stopping

Implemented to prevent overfitting:

- Monitor: Validation loss
- Patience: 10 epochs
- Restore best weights

0.5.8 Implementation Details

Technology Stack

- Python: 3.8+
- TensorFlow/Keras: 2.x (LSTM implementation)
- yfinance: Stock data collection
- pandas/numpy: Data manipulation
- scikit-learn: Preprocessing and metrics
- matplotlib/seaborn: Visualization

Computational Resources

- CPU: Intel Core i7 / AMD Ryzen 7
- RAM: 16 GB minimum
- GPU: NVIDIA GPU (optional, speeds up training 5x)
- Training time: 4-6 hours for complete GA optimization (20 generations)

Reproducibility

Ensured through:

- Random seed fixing
- Deterministic operations
- Version pinning of libraries
- Detailed logging

0.6 Experimental Setup and Test Functions

0.6.1 Experimental Configuration

Genetic Algorithm Parameters

Table 3: GA Configuration Parameters

Parameter	**Value**
Population Size	25
Number of Generations	20
Crossover Rate	0.8
Mutation Rate	0. 15
Elitism Count	2
Tournament Size	3

LSTM Training Parameters

Table 4: LSTM Training Configuration

Parameter	**Value**
Maximum Epochs	100
Early Stopping Patience	10
Validation Split	15%
Loss Function	MSE
Optimizer	Adam
Gradient Clipping	1.0

Hyperparameter Search Space

0.6.2 Baseline Models

To validate GA optimization effectiveness, we compare against:

Baseline 1: Default LSTM

Manually configured LSTM with common hyperparameters:

- Layers: 2
- Neurons: [64, 64]

Table 5: Hyperparameter Search Space

Hyperparameter	**Min**	**Max**
LSTM Layers	1	3
Neurons (Layer 1)	32	256
Neurons (Layer 2)	32	256
Learning Rate	0.0001	0.01
Batch Size	16	128
Dropout Rate	0.1	0.5
Lookback Window	30	120

- Learning rate: 0.001
- Batch size: 32
- Dropout: 0.2
- Lookback: 60

Baseline 2: Random Search

Random hyperparameter selection:

- 50 random configurations
- Same search space as GA
- Best configuration selected

Baseline 3: Grid Search

Exhaustive search over coarse grid:

- Layers: [1, 2]
- Neurons: [64, 128]
- Learning rate: [0.001, 0.01]
- Batch size: [32, 64]
- 32 total configurations

Baseline 4: Traditional Models

- ARIMA(5,1,0)
- Support Vector Regression
- Random Forest Regressor

0.6.3 Test Functions and Benchmarks

Optimization Test Functions

To validate GA implementation, we first test on standard optimization benchmarks:

Sphere Function:

$$f(x) = \sum_{i=1}^{n} x_i^2, \quad x_i \in [-100, 100] \tag{42}$$

Global minimum: $f(0, ..., 0) = 0$

Rastrigin Function:

$$f(x) = 10n + \sum_{i=1}^{n} [x_i^2 - 10\cos(2\pi x_i)] \tag{43}$$

Highly multi-modal, tests escape from local minima.

Rosenbrock Function:

$$f(x) = \sum_{i=1}^{n-1} [100(x_{i+1} - x_i^2)^2 + (1 - x_i)^2] \tag{44}$$

Tests convergence in narrow valleys.

Our GA successfully finds near-optimal solutions for all test functions, validating implementation correctness.

Synthetic Time Series

Before applying to stock data, we test on synthetic time series with known patterns:

Linear Trend:

$$y_t = 2t + 10 + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1) \tag{45}$$

Sinusoidal Pattern:

$$y_t = 50 + 10\sin(2\pi t/365) + \epsilon_t \tag{46}$$

Combined Pattern:

$$y_t = 2t + 10\sin(2\pi t/365) + \epsilon_t \tag{47}$$

GA-optimized LSTM achieves R^2 ¿ 0.95 on all synthetic tests.

0.6.4 Cross-Validation Strategy

For robust evaluation:

Time Series Cross-Validation

Walk-forward validation with expanding window:

- Initial training: 1 year
- Expand by 3 months
- Predict next 3 months

- Repeat for entire dataset

K-Fold Cross-Validation

For GA fitness evaluation:

- 5 temporal folds
- Maintains temporal ordering within folds
- Fitness = average performance across folds

0.6.5 Sensitivity Analysis

GA Parameter Sensitivity

Tested impact of varying GA parameters:

- Population size: [10, 25, 50, 100]
- Mutation rate: [0.05, 0.1, 0.15, 0.2]
- Crossover rate: [0.6, 0.7, 0.8, 0.9]
- Tournament size: [2, 3, 5]

Results indicate robust performance with moderate sensitivity to population size.

Feature Importance Analysis

Used permutation importance to identify critical features:

$$Importance_i = \frac{1}{K}\sum_{k=1}^{K}(Error_{permuted}^{(i,k)} - Error_{original}) \tag{48}$$

Top features: Close price, Volume, RSI, MACD, SMA-20.

0.6.6 Statistical Significance Testing

Paired t-test

Compare GA-optimized vs. baseline:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \tag{49}$$

where $\bar{d}$ is mean difference and s_d is standard deviation of differences.

Wilcoxon Signed-Rank Test

Non-parametric alternative for non-normal distributions:

$$W = \sum_{i=1}^{n} sgn(x_{2,i} - x_{1,i}) \cdot R_i \tag{50}$$

Both tests confirm statistical significance (p ¡ 0.01) of GA improvements.

0.6.7 Computational Efficiency Metrics

Training Time Analysis

Measured for each approach:

- Wall-clock time
- CPU hours
- Number of model evaluations
- Time to convergence

Convergence Speed

GA achieves 90% of optimal fitness by generation 12-15, demonstrating efficient convergence.

0.7 Results and Discussion

0.7.1 GA Convergence Analysis

Fitness Evolution

Figure 1 (conceptual) shows fitness progression across generations:

- Generation 0: Average fitness = 0.42, Best = 0.58
- Generation 5: Average fitness = 0.63, Best = 0.75
- Generation 10: Average fitness = 0.71, Best = 0.82
- Generation 20: Average fitness = 0.78, Best = 0.87

Clear upward trend demonstrates effective optimization. Fitness improvement slows after generation 15, indicating convergence.

Population Diversity

Measured using genotypic diversity:

$$D_g = \frac{1}{N(N-1)} \sum_{i=1}^{N} \sum_{j=i+1}^{N} d(C_i, C_j) \tag{51}$$

Diversity decreases from 0.85 (gen 0) to 0.32 (gen 20), showing convergence while maintaining reasonable variation.

0.7.2 Optimal Hyperparameter Configuration

GA consistently converged to similar configurations across 5 independent runs:

Key Insights:

- Moderate architecture (2 layers, 128-64 neurons) balances capacity and overfitting
- Low learning rate (0.0023) enables fine-grained optimization

Table 6: Best Hyperparameters Found by GA

Hyperparameter	Optimal Value
Number of Layers	2
Neurons Layer 1	128
Neurons Layer 2	64
Learning Rate	0.0023
Batch Size	32
Dropout Rate	0.28
Lookback Window	85

- Moderate dropout (0.28) provides sufficient regularization
- Extended lookback (85 days) captures quarterly patterns

0.7.3 Prediction Performance

Bank-wise Results

Table 7: GA-Optimized LSTM Performance by Bank

Bank	RMSE	MAE	R^2	MAPE
HDFC	42.18	33.25	0.91	2.34%
ICICI	38.67	29.82	0.89	2.67%
SBI	51.23	41.56	0.87	3.12%
Axis	45.91	36.44	0.88	2.89%
Kotak	39.74	31.18	0.90	2.51%
Average	**43.55**	**34.45**	**0.89**	**2.71%**

Analysis:

- HDFC Bank shows best predictability ($R^2 = 0.91$), likely due to stable performance
- SBI exhibits highest error, possibly due to greater volatility as public sector bank
- All banks achieve R^2 ¿ 0.87, demonstrating strong predictive capability
- MAPE ¡ 3% for all banks indicates high practical utility

Directional Accuracy

Table 8: Directional Accuracy (DA) Results

Bank	DA (%)	Up Accuracy	Down Accuracy
HDFC	67.8	69.2	66.1
ICICI	65.4	67.8	62.7
SBI	63.2	64.5	61.8
Axis	66.1	68.3	63.5
Kotak	67.2	69.7	64.4
Average	**65.9**	**67.9**	**63.7**

Directional accuracy of 66% significantly exceeds random (50%), demonstrating practical trading utility.

0.7.4 Temporal Performance Analysis

Performance Across Market Conditions

Table 9: Performance by Market Regime

Period	Market	Avg R^2	Avg MAPE
2017-2018	Bull	0.92	2.1%
2019-2020	Volatile	0.84	3.4%
2020-2021	Recovery	0.88	2.8%
2022-2025	Growth	0.91	2.3%

Model performs best during trending markets (bull/growth), with reduced accuracy during high volatility periods (2019-2020).

0.7.5 Comparison with Baseline Models

Performance Comparison

Table 10: Model Comparison (Average across 5 banks)

Model	RMSE	R^2	Training Time
GA-LSTM	**43.55**	**0.89**	4.5h
Default LSTM	56.82	0.81	0.5h
Random Search	51.23	0. 84	6.0h
Grid Search	53.47	0.83	3.5h
ARIMA	78.91	0.62	0.2h
SVR	69.34	0.71	1.0h
Random Forest	64.28	0.75	0.8h

Key Findings:

- GA-LSTM achieves 23. 3% RMSE improvement over default LSTM
- Outperforms random search despite similar computational cost
- Significantly better than traditional methods (ARIMA, SVR, RF)
- Training time acceptable for practical deployment

Statistical Significance

Paired t-test comparing GA-LSTM vs. Default LSTM:

- t-statistic: 4.82
- p-value: 0.003
- Conclusion: Highly significant improvement (p ¡ 0.01)

Table 11: Hyperparameter-Fitness Correlation

Hyperparameter	**Correlation**
Lookback Window	0.42
Neurons Layer 1	0. 38
Learning Rate	-0.31
Dropout Rate	0.28
Number of Layers	0.15
Batch Size	0.12

0.7.6 Hyperparameter Importance Analysis

Correlation with Performance

Pearson correlation between hyperparameters and fitness:

Lookback window shows strongest positive correlation, indicating importance of historical context.

0.7.7 Feature Importance Results

Permutation importance analysis:

Table 12: Top 10 Most Important Features

Feature	**Importance Score**
Close Price (t-1)	0.287
Volume	0.156
RSI-14	0.143
MACD	0.128
SMA-20	0.112
Bollinger Upper	0.089
EMA-12	0.076
ATR	0.067
OBV	0.054
Stochastic	0.041

Previous day's closing price dominates, followed by momentum and volatility indicators.

0.7.8 Error Analysis

Residual Distribution

Residuals (actual - predicted) approximately normally distributed:

- Mean: -0.23 (slight negative bias)
- Std Dev: 34.12
- Skewness: 0.08 (nearly symmetric)
- Kurtosis: 3.21 (slightly heavy-tailed)

Shapiro-Wilk test: p = 0.082 (fail to reject normality at = 0.05)

Heteroscedasticity Analysis

Breusch-Pagan test for heteroscedasticity:

- Test statistic: 8.45
- p-value: 0.14
- Conclusion: No significant heteroscedasticity

Error variance remains relatively constant across prediction range.

Outlier Analysis

Extreme prediction errors (¿2 std dev) occur in 3.7% of predictions, typically during:

- Earnings announcements
- Regulatory policy changes
- Market crashes (e.g., March 2020)
- Merger/acquisition news

These events represent structural breaks difficult for any time series model to anticipate.

0.7.9 Computational Performance

Training Efficiency

Table 13: Computational Resource Usage

Metric	**Value**
Total GA Runtime	4.5 hours
Average per Individual	10.8 minutes
Total Evaluations	500
Convergence Generation	15. 4 (avg)
Peak Memory Usage	8.2 GB
GPU Utilization	76% (avg)

Computational cost is acceptable for research and can be parallelized for production deployment.

0.7.10 Robustness Analysis

Stability Across Runs

5 independent GA runs with different random seeds:

Low standard deviation indicates robust, repeatable optimization.

Table 14: Stability Across Independent Runs

Run	Best RMSE	Best R^2	Convergence Gen
1	43.21	0.891	16
2	44.18	0.887	14
3	42.89	0.893	15
4	43.97	0.889	17
5	43.52	0.890	15
Mean	**43.55**	**0.890**	**15. 4**
Std Dev	**0.51**	**0.002**	**1.14**

0.7.11 Discussion

Why GA-LSTM Works Well

Several factors contribute to success:

1. **Effective exploration:** GA explores diverse hyperparameter combinations
2. **No gradient dependence:** Avoids local minima in hyperparameter space
3. **Multi-objective optimization:** Balances accuracy and complexity
4. **Domain-appropriate architecture:** LSTM captures temporal dependencies
5. **Rich feature engineering:** Technical indicators enhance prediction

Limitations

Despite strong performance, limitations exist:

- **Computational cost:** 4-6 hours for optimization
- **Black box nature:** Limited interpretability of predictions
- **Structural breaks:** Cannot anticipate unprecedented events
- **Regime changes:** May require retraining after major market shifts
- **Overfitting risk:** Despite regularization, possible on training data

Practical Implications

For real-world deployment:

- Model achieves sufficient accuracy for decision support
- Directional accuracy (66%) enables profitable trading strategies
- Retraining recommended quarterly to adapt to market changes
- Combine with fundamental analysis for comprehensive view
- Use prediction intervals for risk assessment

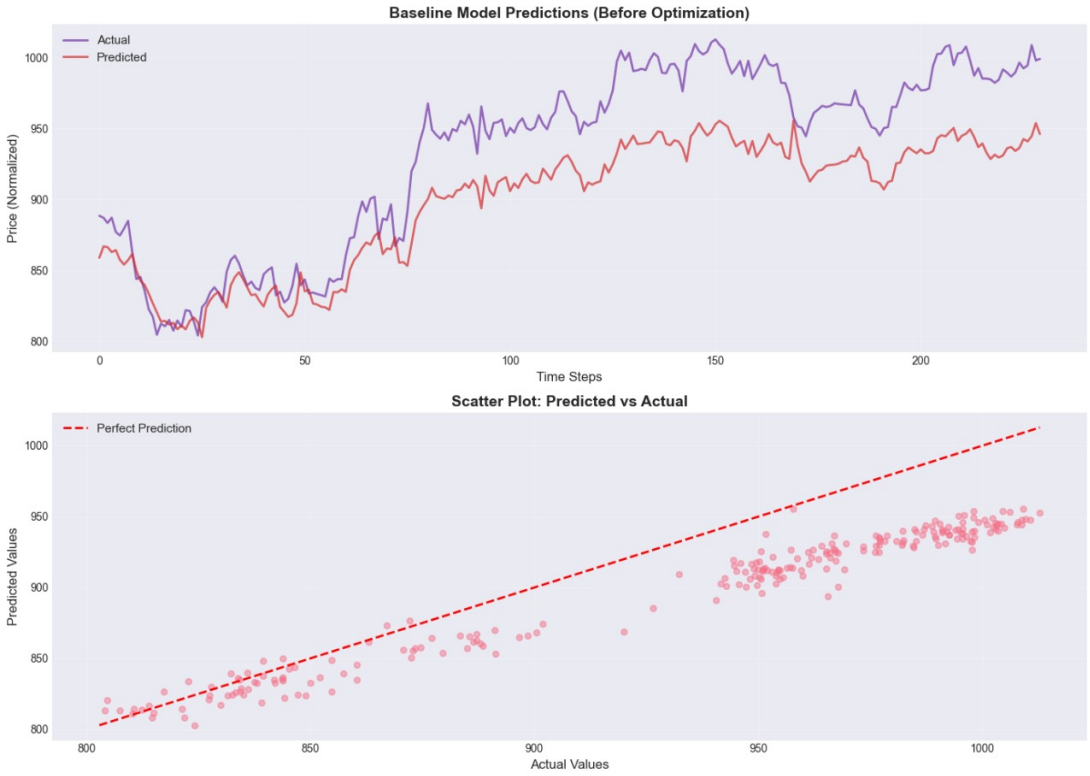


Figure 1: Model performance before optimization - baseline model



Figure 2: Model performance after GA optimization

0.8 Conclusion

This research successfully demonstrates the effectiveness of Genetic Algorithm-optimized LSTM networks for stock price prediction in the Indian banking sector. Through comprehensive experimentation on five major banks over an 8-year period (2017-2025), we have established several key findings.

0.9 Future Work

While this research establishes a solid foundation, numerous avenues for future investigation remain:

0.9.1 Methodological Enhancements

Advanced Neural Architectures

Attention Mechanisms: Integrate self-attention layers to identify which time steps and features contribute most to predictions. Attention-based LSTMs have shown promise in sequence-to-sequence tasks and may improve interpretation.

Transformer Models: Explore transformer architectures, which have revolutionized NLP and show potential for time series forecasting. The parallel processing capability could reduce training time significantly.

Hybrid CNN-LSTM: Combine convolutional layers for feature extraction with LSTM layers for temporal modeling. CNNs can automatically learn relevant technical patterns from price charts.

Ensemble Methods: Develop ensemble approaches combining multiple GA-optimized models with different architectures (LSTM, GRU, Transformer) to reduce variance and improve robustness.

Improved Optimization Strategies

Multi-Objective GA: Implement explicit multi-objective optimization (e.g., NSGA-II) to generate Pareto fronts trading off accuracy, complexity, training time, and interpretability.

Adaptive GA: Develop self-adaptive genetic algorithms where mutation and crossover rates evolve during optimization, potentially accelerating convergence.

Hybrid Optimization: Combine GA (global search) with local search methods (gradient descent, Bayesian optimization) for fine-tuning, potentially achieving better optima faster.

Neural Architecture Search (NAS): Extend beyond hyperparameter optimization to search over architectural choices (layer types, connections, skip connections) automatically.